



BUDDHA'S BRAIN

The Practical Neuroscience of
Happiness, Love & Wisdom

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The Big “So What” ???

In this book, the authors share the latest neuroscience discoveries that explain **how our minds work** and **how 2,000 years of Buddhist contemplative practices** can help to awaken our minds and create **positive change**.

Introduction

Our brains are made of 1.1 trillion cells, including 100 billion neurons, which are constantly transmitting signals and information via synapses. Science has shown that our brains continue to change throughout our lives.

 Every thought and feeling that we have generates mental activity. Our mind is the combination of all the energy and information flowing through our neural circuits. These affect our neural structures to shape our brain.

 Our brain regulates complex functions in the body, which in turn allow us to interact with the outside world (e.g. by using our senses, arms or legs). It affects what we perceive to influence our mental activities.

 In short, the mind and brain are co-dependent, and **we can deliberately shape our brain** by managing our mental activities.

This book is structured like an operating manual for your brain. The authors present a mix of:

 **Scientific background**, including the parts of the brain, how they work, and the body's biochemical processes; and

 **Practical tips and exercises** that are scientifically proven to bring positive change.

KEY QUOTES

“Buddha’s Brain is an invitation to use the focus of your mind to harness the power of attention to enhance your life and your relationships with others.”

“We can actually use the mind to change the brain.”

“You can do small things inside your mind that will lead to big changes in your brain and your experience of living.”

“This book is about how to reach inside your own brain to create more happiness, love, and wisdom.”

“Learn both brain science and practical inner ways to enhance well-being, develop ease and compassion, and reduce suffering.”

In this summary, we'll start by explaining why we suffer (from both Buddhist and scientific perspectives), before moving into specific practices to shape our minds and brains for greater happiness, love and wisdom.

Part 1: Why we suffer

“Suffering” in this book refers to our feelings of frustration, stress, hurt, loneliness etc. In Buddhism, suffering is said to come from 3 poisons: *greed* (chasing after rewards), *hatred* (avoiding threats) and *delusion* (of how things really are). Science helps us to understand how such feelings arise, and how we can reduce the suffering.

Evolution and Suffering

One reason for our suffering is the conflict between our survival strategies and how the world works.



Mankind has evolved with 3 key survival strategies:

- We create a mental division between ourselves and the world, and between different mental states;
- We seek stability; and
- We seek opportunities and avoid threats to our survival and reproduction.



However, the world often works in the opposite way:

- Everything's connected, from how our body parts work together, to how we interact with the people and elements in our environment (like the air and water we need for survival). We suffer when we try to isolate what's fundamentally connected.
- Everything keeps changing, from our body systems (e.g. temperature, neural synapses, heart beat) to the shifting clouds and rivers. When we seek predictability and control in an everchanging world, it creates stress and frustration.

KEY QUOTES

“When you understand why you feel nervous, annoyed, hassled, driven, blue or inadequate, those feelings have less power over you.”

“Since we are each connected and interdependent with the world, our attempts to be separate and independent are regularly frustrated.”

- We can't satisfy all opportunities and remove all threats.

i. When we *experience rewards*, dopamine and “pleasure chemicals” are released in our bodies and we seek out more of the rewards. We suffer because (a) it's unpleasant to crave for something, (b) we feel disappointed when our expectations aren't met or when we get decreasing satisfaction from the same reward, and (c) all experiences eventually end.

ii. We also feel anxious because we have a *negativity bias*. Our brain (a) constantly scans the environment for threats, (b) is more sensitive to negative inputs than positive inputs, and (c) pulls up negative memories whenever we encounter similar triggers. This creates a vicious cycle of pessimism and negative experiences.



The conflict above is amplified by our mental simulator. Our brain creates its own virtual reality, filling information gaps and simulating the future, all without our awareness. These mini-movies in our minds:

- Take us away from the present;
- Create assumptions and limiting beliefs about how things will turn out; and
- Reinforce negative experiences which may not actually happen.



Self-Compassion is a key ingredient to help us cope with difficult situations, build resilience and self-worth. You can nurture self-compassion by strengthening its related neural circuits:

- Activate the circuits in your brain by thinking of (a) how it feels to receive care from someone who loves you, and (b) how you feel toward a child or someone you love.
- Think of your own suffering and extend the same compassion to yourself. Allow the sensation to fill your whole body. Heighten the feeling by putting your palm on your cheek/heart and saying, “May I be happy again. May the pain of this moment pass.”

KEY QUOTES

“Everything that begins must also cease. Everything that comes together must also disperse. Experiences are thus incapable of being completely satisfying.”

“The simulator takes you out of the present moment and sets you chasing after carrots that aren't really so great while ignoring more important rewards (such as contentment and inner peace).”

The First and Second Darts



All of us experience physical pain, which is the “first dart” in Buddhist terms. However, **most of our suffering comes from the “second darts” that we throw at ourselves.**

- For example, if you trip and fall, you’ll feel the first darts of pain. If you get angry at yourself for being clumsy or upset at your spouse for not helping you, you’re throwing second darts.
- Most second-dart reactions occur without first darts, i.e. we create our own suffering, even in positive situations, e.g. you receive a birthday gift from a colleague but end up feeling guilty that you didn’t give her one for her birthday. Each dart you throw can trigger even more darts.



Overheating the system. Suffering is a tangible phenomenon that can be observed and felt in your body, through your Sympathetic Nervous System (SNS) and the Hypothalamic-Pituitary-Adrenal axis (HPAA) of the endocrine system.

- When faced with a threat (e.g. a bike is speeding toward you), your body naturally snaps into red alert. Your body releases signals, chemicals, and stress-related hormones. You become fully alert and *ready for action*, with increased heart rate, dilated pupils, suppressed reproduction and digestion systems, etc.
- Such reactions help us to deal with tough or dangerous situations. However, second darts can put us on red-alert all the time, which is bad for physical and mental wellness. Constant stress causes both *physical issues* (like ulcers, constipation, heart attacks, lowered libido) and *mental issues* (like anxiety and depressed moods).



Cooling the system. The Parasympathetic Nervous System (PNS) is your normal resting state; it helps you to stay calm and clear-headed, and quietens the mind to support contemplative insight. The SNS is like the accelerator of your car while the PNS is like the brakes—ideally, the 2 systems should be balanced like this:

- PNS is activated as your baseline, offering ease and peace;

KEY QUOTES

“As long as you live and love, some of those darts will come your way.”

“Suffering is not abstract or conceptual. It’s embodied: you feel it in your body, and it proceeds through bodily mechanisms.”

“Even if your pot isn’t boiling over, just simmering along with second-dart activation is quite unhealthy.”

“Suffering has clear causes in your brain and body, so if you change its causes, you’ll suffer a lot less.”

- SNS is mildly activated for vitality and enthusiasm; and
- There are occasional SNS spikes for challenging situations.

 **Practicing awareness.** You can shape your mental and physical reactions by becoming aware of your suffering as it arises. Observe your feelings and responses, so you can regulate your impulses until you no longer throw second darts and react to the suffering. For the rest of this summary, we'll look at how you can manage your mental activities to shape your neural connections and your brain for greater happiness, love, and wisdom.

Part 2: Happiness

Since our brains have a negativity bias, we tend to accumulate negative experiences faster than positive ones. **The key to happiness is to boost the positive experiences, so they can take root.**

Absorb the Positives



Positive emotions bring physical and mental health benefits, e.g. stronger immune system, stress-resilient cardiovascular system, etc.



To internalize the positives, use these 3 steps:

- Look out for positive facts (e.g. an act of kindness, a blooming flower) and convert them into positive experiences (by focusing on how they affect you).
- Savor the experiences, intensifying the sensations to fortify the memory thread. Focus on the rewarding aspects of the experience to increase dopamine release for positive neural associations.
- Imagine the experience (thoughts, emotions, sensations) being deeply absorbed into your mind and body.



Each time your brain calls up a memory, it modifies it by filling gaps and reinterpreting details, all without your conscious realization. When you hold 2 things in your mind at the same time, your brain also forms new neural connections. Thus, **you can rebuild your memories to heal pain, using these tips:**

KEY QUOTES

"As the saying goes pain is inevitable but suffering is optional. If you can simply stay present with whatever is arising in awareness... without reacting further, then you will break the chain of suffering right there."

"Much as your body is built from the foods you eat, your mind is built from the experiences you have."

"Positive experiences can also be used to soothe, balance, and even replace negative ones."

- Deliberately call up positive thoughts/feelings while recalling a painful memory, so you gradually dull the old pain.
- Take notice of your recurring upsets to identify the root causes. Use positive experiences to treat the causes of the old pains. In addition, each time the negative material surfaces, neutralize it with positive emotions.

By using just a few seconds each time, you can build new, positive structures in your brain to alter your thoughts and actions.

Calm Yourself



You can stimulate the PNS to calm your body, brain, and mind.



To activate the PNS:

- Practice relaxation. Scan your body mentally and progressively relax each part of your body, e.g. your eyes, jaw, left foot. You can also use *imagery* to help yourself relax e.g. imagine sitting by a peaceful lake.
- Practice breathing. Try breathing from your *diaphragm* for 1-2 minutes. Or, take a *deep breath*, hold it for a few seconds and exhale slowly as you relax. You can also balance your heartbeat: inhale and exhale evenly, and imagine the breath going through your heart area alongside feelings like gratitude or love.
- Adopt a daily meditation routine, no matter how brief. Here's an example of a mindfulness meditation:
 - Find a comfortable place where you won't be disturbed. Breathe deeply and relax. Put down your burdens like putting down a heavy bag, knowing you can pick up what you want after the session.
 - Be aware of sounds around you, without thinking about them. Focus on your breathing, feeling the cool air entering and warm air leaving, and your chest/belly rising and falling.
 - If your mind wanders, simply bring it back to your breathing. Anchor yourself in your breath and allow all thoughts,

KEY QUOTES

"The art is to find a balance in which you remain mindful, accepting and curious regarding difficult experiences—while also taking in supportive feelings and thoughts."

feelings, images etc. to pass through. Stay with the feeling of peace, space and awareness.



Although you can never be 100% safe, you can soothe your fears and avoid overreacting to threats.

- Relax your body using the methods above.
- Connect with nurturing and supportive people in person or via visualization.
- Visualize protective figures (e.g. your parents, a guardian angel), or being surrounded by a protective force field. Pay extra attention when people care for and stand up for you, and carry the feeling with you.
- Evaluate your fears realistically. You may realize they're not as likely or as bad as you'd feared. Focus on what you can do and the resources you can tap on. When fear surfaces, simply observe what's happening and let the feeling pass without being alarmed.



Seek refuge—be it people, activities, places, or thoughts—that energize you and make you feel safe. This is where you can let down your guard, learn freely and refuel yourself to face challenges. *Make a list* of as many of your refuges as possible. Think about them and get familiar with the *associated feelings and sensations*.

Fortify your Intentions



It's possible to tap on your mental strength to improve happiness.



Understand your motivation hubs.

- There are 2 key motivation hubs in your brain:
 - i. *The amygdala hub* is more primal in nature. It constantly scans what's happening around you to identify opportunities and threats, providing information for decision-making.
 - ii. The *Anterior Cingulate Cortex (ACC) hub* holds your

KEY QUOTES

"Deliberately feeling safer helps control the hardwired tendency to look for and overreact to threats."

"In effect, you are giving yourself today the attention and attunement you should have received as a child."

"Keep separating from the fear; settle back into the vast space of awareness through which fear passes like a cloud."

"The amygdala is at the center of bottom-up, reactive, distributed, passionate motivation."

"The ACC is at the center of top-down, deliberate, centralized, reasoned motivation."

“working memory”, where your brain places information to solve problems and make decisions. The ACC sets your intentions, decides where you focus your attention, and how you pursue your goals and deal with conflicts.

- The 2 hubs influence each other in a feedback loop. When they are in conflict, you feel as if your head and heart are pulling in different directions. When they are aligned, your motivation is the strongest.
- Intention is a form of desire, and desires need not cause suffering. Your intentions only lead to suffering when (a) it involves craving, i.e. you’ll be upset if you don’t get something, and (b) the desire is for harmful purposes, e.g. the 3 poisons.



Strong intentions are associated with energy and determination:

- Bring up the feeling of determination by quickening your breathing slightly or tightening your shoulders as if you’re going to carry a heavy load. Use facial expressions to intensify your emotions.
- Reinforce associated neural pathways by visualizing a time you felt really strong, and observe how your body feels. Carry this feeling with you, and call it up when dealing with challenges. Reinforce it with words like “I’m feeling strong”, “I have the right to be strong.”

Seek Equanimity



Equanimity means staying centered, so you do not react to your reactions.



Understand that both pain and pleasure are transient. Let them pass and focus on what you can do now to shape future outcomes. Learn to accept and be at peace with whatever you feel—whether it’s pleasant or unpleasant—so you create a buffer between your triggers and suffering. When you no longer react, identify with or own your pain and pleasure, you feel a sense of peace and freedom.



The equanimous state of mind can be cultivated through 4 neural conditions.

KEY QUOTES

“Desire per se is not the root of suffering; craving is.”

“When you are equanimous, you don’t chase after enjoyable experiences or resist negative ones.”

“Neither pleasure nor pain is worth claiming as your own or identifying with.”

“Over time, equanimity deepens into a profound inner stillness that is a defining characteristic of contemplative absorption.”

- Stay “disenchanted”, so you’re not disappointed or dissatisfied with life. Set the intention not to react to your reactions.
- Cultivate a steady mind, by being aware of passing thoughts and feelings without being attached to them. Pay attention to how you feel/think when your mind is still and neutral, and bring yourself back to this state whenever you feel agitated.
- Develop spacious consciousness, and treat your fleeting thoughts and emotions like shooting stars in an infinite space.
- Make a list of situations that trigger big reactions in yourself, and rate their intensity. Use the PNS-activation approaches earlier to practice tranquillity, starting from the least intense situations.

Part 3: Love

The 2 Wolves in your Heart

We all have 2 wolves inside us: one of love and one of hate. Whichever wolf we feed will grow stronger.

- Love. We’re evolved to bond emotionally with significant people like our family members. Love feels good due to associated chemicals like oxytocin and dopamine. Losing love feels bad as it activates neural systems for physical pain.
- Hate also evolved out of the need for protection against “outsiders”. We tend to feel loyalty for “us” but aggression for “them”. We can’t kill the wolf of hate, but we can leash it and feed the wolf of love.

Compassion and Assertion

To nurture the wolf of love, we need empathy, compassion (to offer care and understanding) and assertion (to address your own needs).

 Empathy is about understanding someone. It’s a foundation to any meaningful relationship. Humans are capable of empathy

KEY QUOTES

“The space of awareness allows every content of mind to be or not to be, to come and to go.”

“Although the wolf of hate gets more headlines, the wolf of love has been painstakingly bred by evolution to be more powerful.”

“The wolf of love sees a vast horizon, with all things included in the circle of ‘us’. That circle shrinks down for the wolf of hate.”

“There is no killing the wolf of hate... But you can keep that wolf carefully tethered.”

because of neural systems that allow us to simulate others' actions, emotions and thoughts.

- When we see others do something, our brain's perceptual motor systems light up as if we're doing those actions ourselves;
- When others experience strong emotions (e.g. fear or anger), it triggers similar neural connections in us. The more we're aware of our inner states, the more we can relate to the feelings.
- We observe others' actions, emotions and thoughts, make an educated guess of what they're thinking, then test it on ourselves to understand where they're coming from.

To build empathy:

- Set the conscious intention to understand the other person. Relax and open up to the person while maintaining a safe space.
- Pay attention to the other person's actions (e.g. movements, posture), feelings (both verbal and physical expressions) and thoughts (e.g. background, motivation, and desires).
- Check back to ensure you've understood them correctly, e.g. "It sounds like....is that right?" or "I get the sense that"
- Be willing to request for and receive empathy.



Cultivate compassion by calling up feelings of love and gratitude before focusing on the person/problem, mentally wishing the other party well, and practicing compassion with 5 different types of people daily (e.g. a friend, a stranger, etc.).



Assert yourself by staying true to your goals and truths.

- Living with unilateral virtue means acting according to your principles and goodness, rather than responding to what others say/do. Write down your personal code of virtues to guide your actions (e.g. "listen more, talk less" or "keep my promises").

KEY QUOTES

"Compassion widens the circle of 'us' while assertion protects and supports everyone inside it."

"Virtue sounds lofty, but it's actually down to earth. It simply means living from your innate goodness, guided by principle."

- To reach an equilibrium, you must satisfy both your head (logically) and your heart (emotionally).

Loving Kindness



Compassion usually relates to others' suffering, whereas kindness (wishing others well) can apply to all contexts.



You can wish others well either generally (e.g. for health or happiness) or specifically (e.g. to secure a desired job or be at peace with a painful loss).

- Do a loving-kindness meditation by modifying your usual meditation to focus on a kind wish with each breath.
- Offer kindness daily through your intentions, words and actions.



Your ultimate test will be to show kindness to someone who has treated you badly. To drop ill-will and cultivate goodwill:

- Look for things that bring you positive emotions, while diffusing negative triggers and feelings as early as possible.
- Avoid arguments by separating your thoughts from others', e.g. imagine your conflicts are separated by a waterfall.
- Don't assume negative intentions by others. Consider the possible reasons they may have acted in a hurtful way.



Expand your "us" circle to eventually include the whole world.

- Redefine your "us" categories. Focus on *similarities* rather than differences, and challenge how you've defined your "not me" categories (e.g. different interests, religions).
- Reduce the sense of threat by asking if others will really harm you. Look for mutual benefit or opportunities for collaboration.
- Practice the loving-kindness meditation. Focus on the feeling of being loved. Allow that feeling to flow through you, toward the people closest to you, then to acquaintances

KEY QUOTES

"If compassion is the wish that beings not suffer, kindness is the wish that they be happy."

"Goodwill and ill will are about intention."

and strangers, until it covers all other living beings on the planet.

Part 4: Wisdom

Mindfulness is the ability to direct your attention at will. It determines what flows through your mind and thus how you shape your brain.

Train your Mind

 Our brain is constantly balancing **3 key needs**: keeping information in awareness, updating awareness with new information, and seeking the right level of stimulation (e.g. food, mates, resources). Imagine information is being held in a working space in your brain. When the contents are stimulating, dopamine is released and the door to the space is closed, allowing you to focus. If the stimulation drops, dopamine levels fall and the door opens to new inputs.

 You can train yourself to be more mindful:

- Before starting any activity that requires focus, set the intention to be mindful. Think of someone who's extremely focused and simulate the sensations and state of mind.
- To cultivate daily mindfulness, slow down, talk less, single-task instead of multi-task, stay centred and focus on breathing during daily activities. Reflect more and simplify your life.
- Stay alert by getting sufficient rest, sitting in an erect position, visualizing light to brighten your mind, and breathing deeply to oxygenate the brain.
- Quieten the mind from distracting thoughts. Suppress left brain chatter by developing awareness of your body as a whole. Instruct your brain's verbal centre to be silent, e.g. tell your mind, "shhh, it's not the time to talk right now", or use mantras/prayers to replace mindless chatter.
- Learn the difference between your awareness (which is an infinite, unchanging space) and fleeting "mind objects". Relax, breathe and observe the mind objects passing through

KEY QUOTES

"When your attention is steady, so is your mind."

"Being mindful simply means having good control over your attention."

"Attention is like a spotlight, and what it illuminates streams into your mind and shapes your brain."

"The brain can't be fully attentive unless it's fully awake."

"Notice the spacelike qualities of awareness: it is boundless, still and silent, empty until something appears, vast enough to hold anything, always present and reliable, and never altered by the mindobjects passing through."

your mind like watching a movie. Realize that your likes and dislikes are merely passing objects.

Improve your Concentration



Meditation is powerful because it focuses your attention like a laser beam. It can improve 5 factors that affect your concentration.



Factors 1 & 2: Applied and sustained attention. You *apply attention* when you initially direct attention to an object. You *sustain attention* when you stay focused on the object.

- Practice focused attention by counting your breath from 1-10 (1 set), and gradually increase the number of sets.
- To filter out distractions, start your meditation by noticing the distractions around you. Invite them into your mind, then ask them to go out before refocusing on your object of attention. If you can't remove the distraction, then make it the focus of your meditation.
- To satisfy your mind's desire for stimulation, focus on novel details like the sensations in a body part (e.g. your tongue), a body area (e.g. your chest area), or break up your breathing into parts (e.g. inhale-pause-exhale).



Factors 3 & 4: Rapture and joy: When you feel intense interest in an object (rapture), or feelings of joy, more dopamine is released and you concentrate better.

- Set intention for rapture and joy, pay attention to them and invite them in. Imagine you're breathing in bliss and joy.
- Make rapture and joy the focus of your meditation. Notice the various shades of happiness, contentment, joy or tranquillity.



Factor 5: Singleness of mind. This comes from all 4 factors above. Unify the mind by focusing on the sense of your entire body working as one. Drop the past and future and focus fully on the present. In the book, the authors also take us through a "*concentration meditation*" combining various techniques covered so far.

KEY QUOTES

"Mindfulness brings insight and wisdom—and the best way to increase your mindfulness is through meditation."

"Singleness of mind involves the unification of awareness, grounded in a deepening absorption in the object of attention."

Relax the Sense of Self



All of us have an “apparent self”, which is merely a concept. There’s no permanent concept of “I”. We are constantly changing as new stimuli and synapses flow through our minds, and the “self” is merely a temporary mind-object. We need the self to function in society, e.g. identify people, connect our life experiences and build relationships, but we needn’t be limited by it. **The key is to play the role of self without being attached to it.**



When you hang on to self, you separate yourself from others and identify with (or try to possess) things that inevitably end, bringing yourself suffering. **Relax the sense of self** and flow with life, so you feel a greater happiness and satisfaction.

- o Take your body for a walk. Enjoy the sights, sounds and sensations without relating to “I”. Realize that you don’t need a sense of self to experience all these.
- o Perceive experiences from the body-mind without identifying with them, e.g. observe that “thoughts are surfacing” rather than “I think....”.



Move away from **envy, possessiveness and self-importance**:

- o Give generously to others, by offering your time, money, patience, forgiveness etc. Focus on compassion and kindness.
- o Cultivate healthy humility, by noticing the good things about yourself and nurturing your self-worth, while acknowledging and caring for others. When your heart is full, you won’t feel the constant need to impress others.



Increase your **connection with the world and with life**:

- o Reflect on how your body exchanges energy and matter (e.g. sunlight, food and water) with the world. Think deeply about what you use daily, e.g. how your food is produced, distributed, and returned to the environment.
- o Notice the space around you and how objects fit in as part of a whole. Realize that everything changes, including the

KEY QUOTES

“Try to keep remembering that who you are as a person dynamically intertwined with the world— is more alive, interesting, capable, and remarkable than any self.”

“Paradoxically, the less your ‘I’ is here, the happier you are.”

“It is not so much that we have a self, it’s that we do self-ing.”

“Self is like a knotted fist: when you open the hand to give, there’s no more fist— no more self.”

“Fill the hole in your heart by taking in the good, once brick at a time.”

“Love the person you are, much as you would care about any person dear to you. But don’t love the self or any mere mind-object.”

self. Reflect on how all life forms fit in this world and wish them well.

KEY QUOTES